

# Requirement vs Repo Gap Analysis

## Context

This document compares your target specification (“Enterprise Intelligence & Investment Research Platform”) against the current implementation in this repository.

Assessment date: 2026-05-28

Scope reviewed: `apps/api`, `apps/web`, `apps/admin`, `apps/worker`, `packages/connectors`, `packages/finance`, `tests`, `docs`.

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## Current Standing (High Level)

The repo is a **working Phase-1 foundation** with core scaffolding in place:

- monorepo app structure is present and runnable
- many API domains exist (identity, evidence, entities, graph, search, reports, review, compliance, monitor, skills, finance)
- connectors package exists with source contracts and connector skeletons
- basic web/admin UIs are available
- monitoring/watchlist/portfolio endpoints exist

But the project is still **early implementation**, not yet a full enterprise-grade research platform.

Main reason: many critical modules are implemented as stubs/skeletons, and enterprise-grade controls, data quality, verification depth, and end-to-end production workflows are still incomplete.

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## Module-by-Module Status

Legend: - **Done (Foundation)** = basic functionality exists - **Partial** = meaningful implementation exists but not production-complete - **Not Started** = no real implementation yet

### 1) Enterprise Workspace & Identity

Status: **Partial**

What exists: - `org/workspace/user/auth` basics (`/auth/register`, `/auth/login`, `/orgs`, `/workspaces`) - role and permission seed logic - project/case creation - audit log endpoint

Gaps: - no SSO/OIDC/SAML - no SCIM provisioning - no MFA policy - no robust tenant isolation verification tests - no hardened session/token lifecycle (rotation, revocation, etc.)

## **2) Source Registry & Connector Framework**

Status: **Partial**

What exists: - source registry APIs (/sources/\*) - source runs and status tracking - source contract model + sample YAML contracts - connector SDK (rate limit, retry, checkpoint primitives)

Gaps: - connector execution path is mostly placeholder - limited real ingestion with durable checkpoints - no robust retry/dead-letter/ops dashboard - weak data quality metrics and SLA observability

## **3) Evidence Vault**

Status: **Done (Foundation) / Partial**

What exists: - raw upload endpoint (/evidence/raw) - hash + storage path persistence - EvidenceRef creation and retrieval

Gaps: - no document viewer with span highlighting - no immutable retention/legal-hold controls - no deep citation quality checks across report sections - no full provenance chain enforcement per claim

## **4) U.S. Source Connectors**

Status: **Partial**

What exists: - connector modules for many U.S. domains (SEC, OFAC, FARA, GLEIF, CourtListener, etc.) - basic tests for connector skeleton behavior

Gaps: - many connectors are fixture/sample based, not full production ETL - missing robust normalization mappings per source - missing end-to-end “discover -> fetch -> raw persist -> normalize -> evidence refs -> relationships” at production depth

## **5) Entity Resolution**

Status: **Partial**

What exists: - entity, alias, identifier models/APIs - deterministic identifier match - fuzzy name resolution - merge proposal/approve/unmerge endpoints - resolution queue

Gaps: - simplistic scoring and merge safety logic - no advanced conflict policies (cross-id contradiction matrices) - no analyst-grade merge review UX - weak reversal fidelity (unmerge is currently naive)

## **6) Intelligence Graph**

Status: **Partial**

What exists: - graph expand/path/related endpoints - relationship evidence linking endpoint

Gaps: - graph queries are still basic - no robust timeline semantics - no mature graph visualization UX - no weighted traversal/trust-aware scoring pipeline

## **7) Investor Intelligence**

Status: **Partial**

What exists: - finance APIs: analyze stock, DCF, reverse DCF, comps, fundamentals - finance package with core computations

Gaps: - no full investor memo pipeline with complete sectioning and evidence traceability - no complete market data provider abstraction layer with production adapters - limited advanced analytics coverage

## **8) Finance Skills Gateway**

Status: **Partial**

What exists: - skills registry and run APIs - provider field with “anthropic/internal” routing concept - artifact storage for runs

Gaps: - anthropic path is simulated, not live provider integration - no strict provider-agnostic contract enforcement across adapters - no strong cost accounting/quotas/policies per run

## **9) Report Generation & Review**

Status: **Partial**

What exists: - report/section/bundle/claim models and APIs - claim-evidence linking - basic claim verification states - comments/suggestions/version diff - export stubs (markdown/html/json/csv)

Gaps: - verifier is still simplistic - no rich editor workflow in frontend - no robust contradiction analysis - no full approval gates + policy routing for sensitive outputs

## **10) Alerts, Monitoring, Portfolios**

Status: **Partial**

What exists: - watchlists CRUD basics - portfolio positions + CSV import + exposure endpoint - alert rules/channels - scan/deliver flow (with webhook + stubs)

Gaps: - alerting logic mostly simulated thresholds - delivery channels (email/slack/teams) are stubs - no advanced portfolio risk/factor/regulatory exposure engine

## 11) Compliance & Governance

Status: **Partial** (early)

What exists: - policy records - export approval request/approve endpoints

Gaps: - no field-level restriction enforcement pipeline - no legal hold/retention enforcement - no full reviewer policy workflow by risk tier

## 12) Search & Retrieval

Status: **Partial**

What exists: - SQL-based search endpoints - OpenSearch endpoint stubs (/searchos)

Gaps: - limited ranking quality and hybrid retrieval sophistication - no robust indexing pipeline + relevance tuning - no vector/hybrid evidence ranking

## 13) Frontend UX (Web + Admin)

Status: **Partial**

What exists: - web routes exist (/search, /graph, /stock, /skills, etc.) - admin shell exists with API health check

Gaps: - still mostly thin pages; no full analyst workbench - no complete review workspace UX - no complete case/project collaboration UX

## 14) Testing & Reliability

Status: **Partial** (low maturity)

What exists: - minimal test scaffolding + some connector tests

Gaps: - very limited integration/E2E coverage - no serious load/security/permission boundary testing - no migration strategy hardening and data lifecycle tests

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## Where We Stand Against Your 15-Issue Plan

Approximate progress:

- **Foundation + module scaffolding:** largely done
- **Production-depth implementation:** not done yet

Practical estimate: ~**35-45% complete** against the full product intent in your handbook.

Reason for this estimate: - breadth of modules is good - depth, hardening, and enterprise-grade controls are still pending

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## Highest Priority Next Work (Recommended Order)

1. **Harden evidence and claims pipeline**
    - strict claim-evidence requirements
    - stronger verifier + contradiction logic
    - provenance integrity checks
  2. **Upgrade entity resolution and merge safety**
    - better confidence model
    - conflict blocking rules
    - reversible merge history with full fidelity
  3. **Productionize top connectors (P0 first)**
    - SEC, FEC, LDA, Congress, Federal Register, USAspending, Open-Corporates, GLEIF, OFAC/OpenSanctions
    - real checkpointed ingestion + quality metrics
  4. **Implement real review workspace UX**
    - section editing, claim status overlays, reviewer queues, approvals
  5. **Strengthen compliance/export policy enforcement**
    - risk-tiered approvals
    - export policy checks
    - audit completeness
  6. **Search and graph quality pass**
    - ranking improvements
    - graph relevance scoring
    - better timeline and relationship evidence UX
  7. **Testing hardening**
    - API integration suite
    - permission boundary tests
    - connector contract tests
    - E2E user workflows
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## Suggested Immediate Sprint Plan (Next 2-3 Sprints)

### Sprint A (Evidence + Claims + Review Core)

- enhance claim verifier
- enforce evidence-required state transitions
- add review gate checks before export
- add integration tests for claim lifecycle

### Sprint B (Connector Depth + Entity Resolution)

- productionize 2-3 highest-value connectors (SEC, USAspending, OFAC)
- improve deterministic + fuzzy resolution scoring
- implement safe merge conflict rules + full unmerge restore logic

### Sprint C (Analyst UX + Compliance)

- build review workspace pages in web/admin
  - add export approval UI and compliance checks
  - ship end-to-end report flow with evidence appendix
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### Bottom Line

You already have a strong base architecture and broad module coverage. To match the full handbook vision, the next phase is about **depth and hardening**:

- make ingestion real
- make evidence/claims rigorous
- make review/compliance enforceable
- make workflows complete end-to-end
- add serious test coverage

That is the path from “good prototype foundation” to “enterprise-ready intelligence platform.”